

Coursework Cover Sheet**Section A - To be completed by the student**

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Penalties: No late work will be accepted. If you are unable to submit coursework on time due to extenuating circumstances, you may be eligible for an extension. Please consult the lecturer.	
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Section B - To be completed by the module leader

Intended learning outcomes assessed by this work (5023CMD):

1. Adhere to the social and legal implications of artificial intelligence. (A4, PLO11).
2. Identify the various approaches to knowledge representation in artificial intelligence. (C4, PLO1)
3. Assess problem-solving strategies under uncertain conditions, maintaining control and accuracy in execution. (P5, PLO3)
4. Illustrate an understanding of knowledge-based systems, and their respective architectures. (C4, PLO1)
5. Demonstrate artificial intelligence approaches in practical tasks to determine the most effective solution for an intelligent application. (P5, PLO3)
6. Verify human factors and the impact on Artificial Intelligence. (A5, PLO10)

Marking scheme	Max	Mark
1. Assignment 2	100	
Total	100	
Lecturer's Feedback		
Internal Moderator's Feedback		

Date: 21st June 2026

Table of Contents

List of Figures	ii
List of Diagrams	ii
List of Tables.....	ii
List of Abbreviations.....	ii
1.0 Intelligent System.....	1
1.1 Sources of Uncertainty in Intelligent Traffic Management.....	2
1.2 Reasoning Under Uncertainty: Fuzzy Logic System	4
1.2.1 System Overview	4
1.2.2 Fuzzification Process	5
1.2.3 Justification of Selected Method.....	7
1.3 Roles of Knowledge Bases	8
1.3.1 Knowledge Storage.....	8
1.3.2 Knowledge Update.....	9
1.3.3 Criticality to System Performance	11
2.0 Machine Learning.....	12
2.1 Choice of ML Model	12
2.1.1 Justification of CNN-YOLOv8.....	13
2.2 Training Considerations	14
2.2.1 Environment Setup in Experiment.....	14
2.2.2 Risks of Poor Parameter Choices.....	16
2.3 Human-AI Interaction.....	17
2.3.1 System Communication Results to Users	17
2.3.2 Improving System Performance Through User Feedback.....	18
2.3.3 Risks of Over-Trust and Under-Trust	19
2.4 Importance of the Machine Learning Component	20
3.0 References.....	21
4.0 Appendix	23

List of Figures

Figure 1. Top Eight Malaysia Cities with the Worst Traffic Congestion (Xiang, 2025)	1
Figure 2. Real-time Traffic Monitoring and Vehicle-pedestrian Classification (Nvidia, 2025) 2	
Figure 3. Traffic Congestion in Georgetown, Penang (Focus, 2022)	3
Figure 4. Detection Under Degraded Weather and Low-visibility Conditions.....	12
Figure A.1 Experimental Results for Learning Rate = 0.1	23
Figure A.2 Experimental Results for Learning Rate = 0.03	23
Figure A.3 Experimental Results for Learning Rate = 0.001	23
Figure B.1 Experimental Results for Regularisation Rate = 0	24
Figure B.2 Experimental Results for Regularisation Rate = 0.001	24
Figure B.3 Experimental Results for Regularisation Rate = 0.1	24
Figure C.1 Sigmoid Activation Function Graph	25
Figure C.2 Linear Activation Function Graph	25
Figure C.3 ReLU Activation Function Graph.....	25

List of Diagrams

Diagram 1. Process Flow of the Proposed Intelligent Traffic Management System	4
Diagram 2. Flowchart of the Mamdani Fuzzy Inference System (FIS) Architecture	5
Diagram 3. Intelligent Traffic Congestion Detection Tree-Sequence UML	8
Diagram 4. Confidence-Based Traffic Control Tree-Sequence UML	9

List of Tables

Table 1. Comparison of Experimental Learning Rates	14
Table 2. Comparison of Regularisation Rates.....	15
Table 3. Comparison of Neural Network Activation Function Types.....	16

List of Abbreviations

Abbreviation	Full Term
UNESCO	United Nations Educational, Scientific and Cultural Organization
MBPP	Penang Island City Council (<i>Majlis Bandaraya Pulau Pinang</i>)
IBM	International Business Machines Corporation
IOC	Intelligent Operations Centre
CCTV	Closed-Circuit Television
PSP	Penang Smart Parking
ML	Machine Learning
CoG	Centre of Gravity
FPS	Frames Per Second
GIS	Geographic Information System
CNN	Convolutional Neural Network
YOLOv8	You Only Look Once version 8

1.0 Intelligent System

Application Area: Smart Cities

Penang's capital, Georgetown only covers 113 square miles but attracts over 8 million tourists annually for its UNESCO heritage streets and food culture. The high volume of traffic is due to nearly 90% reliance on private vehicles creates severe congestion. The average travel speed drops to 10 kilometres per 27 minutes and Georgetown is ranked 73rd among the world's most congested cities (The Sun, 2025).

Rank by filter	World rank ▲	City	Average travel time per 10 km ▼	Change from 2023 ▼	Congestion level % ▼	Time lost per year at rush hours ▼	Congestion world rank ▼
1	73	George Town Malaysia	26 min 36 s	+ 10 s	40%	75 hours	45
2	139	Kota Bharu Malaysia	22 min 30 s	- 20 s	34%	60 hours	121
3	164	Ipoh Malaysia	21 min 48 s	+ 10 s	26%	51 hours	306
4	215	Johor Bahru Malaysia	19 min 58 s	no change	24%	49 hours	348
5	285	Kuala Lumpur Malaysia	17 min 26 s	- 10 s	28%	68 hours	258
6	385	Seberang Perai Malaysia	14 min 30 s	no change	28%	49 hours	253
7	421	Klang Malaysia	13 min 9 s	- 10 s	17%	27 hours	448
8	437	Kajang Malaysia	12 min 33 s	+ 10 s	23%	41 hours	362

Figure 1. Top Eight Malaysia Cities with the Worst Traffic Congestion (Xiang, 2025)

The MBPP collaborated with IBM had established an IOC utilizing smart CCTV feeds and data analytics to manage urban traffic (IBM, 2020). The PSP system successfully reduced the parking search time by 50% (Sigfox, 2021). These initiatives provide the practical context for the implementation of the proposed fuzzy logic-based intelligent traffic signal system for Georgetown's congested urban network.

1.1 Sources of Uncertainty in Intelligent Traffic Management

Incomplete Data

Incomplete data often stems from deployment mishaps or authentication failures during system updates. In early 2025, system operator HeiTech Padu Bhd's version 3.0 update for the PSP platform introduced a biometric login glitch that was incompatible with existing user account data (The Star, 2025). During the incident, local councils suspended parking enforcement and waived fees. Such incompleteness reduces the reliability of downstream analytics as ML models trained on fragmented records produce inaccurate behavioural predictions.

Noisy Inputs

Noisy inputs from environmental and sensor interference degrade the classification confidence scores. In Malaysia, tropical heat haze rising off the asphalt disrupts background subtraction, while monsoon downpours blur CCTV lenses and cause lens distortion. This introduces computational noise into computer-vision frameworks, thereby causing vehicle-counting algorithms to experience bounding-box instability, misclassifications or false negatives during peak weather.

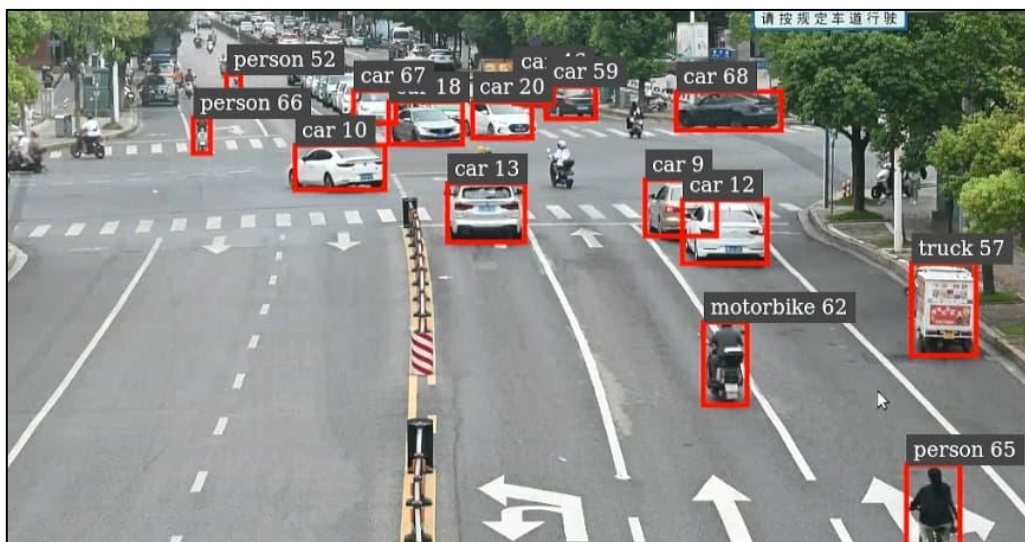


Figure 2. Real-time Traffic Monitoring and Vehicle-pedestrian Classification (Nvidia, 2025)

Changing Environment

A third category arises from changing road topology or operation constraints of the operating space over time. The MBPP frequently alters local traffic rules, such as converting traditional two-way heritage lanes into one-way loops to manage congestion or implementing sudden road closures for cultural processions such as the annual Kek Lok Si festival. These changes are announced on short notice and not immediately reflected in digital maps, which leads to inaccurate prediction of current traffic behaviour.



Figure 3. Traffic Congestion in Georgetown, Penang (Focus, 2022)

Human Unpredictability

Human unpredictability arises from behaviour that deviates from traffic regulations. George Town attracts large numbers of tourists to narrow roads such as Lebuh Chulia and Lebuh Pantai. Tourists stop rental bicycles or trishaws abruptly to photograph murals, while local drivers often double-park temporarily for food pickups or passenger drop-offs. These spontaneous actions create short-term traffic bottlenecks difficult for intelligent traffic-management systems to become computationally uncertain as such stochastic human behaviour cannot be fully represented through static rule-based systems.

1.2 Reasoning Under Uncertainty: Fuzzy Logic System

1.2.1 System Overview

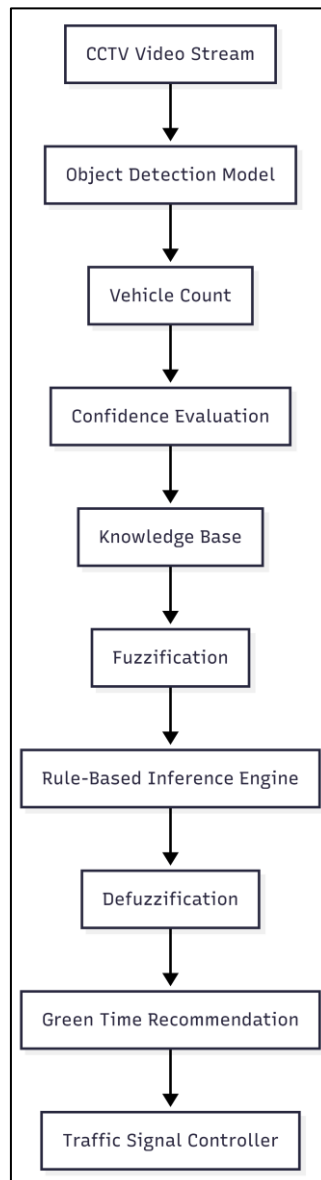


Diagram 1. Process Flow of the Proposed Intelligent Traffic Management System

This Diagram 1 illustrates an intelligent traffic management system that integrates CCTV video streams with an object detection model to track vehicles. The vehicle counts and confidence score feed into a fuzzy logic decision-making core. This core consists of fuzzification, a rule-based inference engine referencing a knowledge base and defuzzification. The system receives input data with uncertainty or imprecision and provides a green time recommendation to the traffic signal controller to optimize signal timing based on real-time traffic conditions.

1.2.2 Fuzzification Process

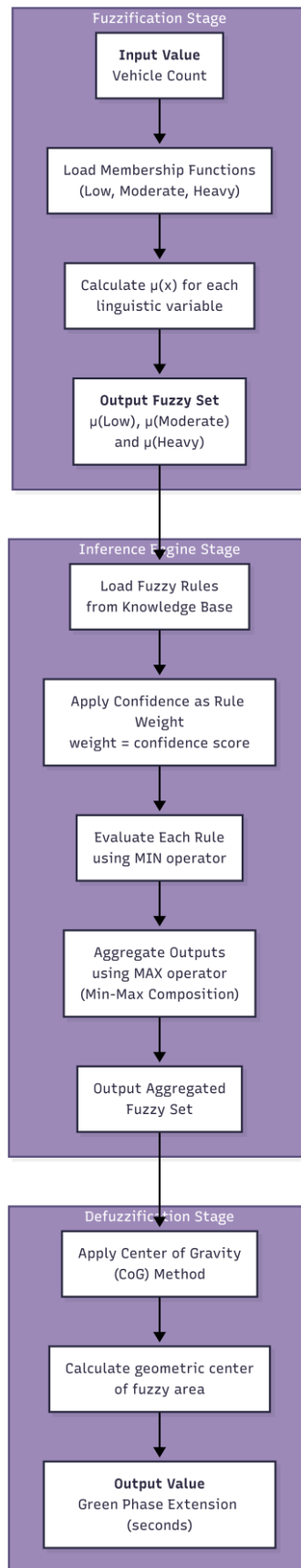


Diagram 2. Flowchart of the Mamdani Fuzzy Inference System (FIS) Architecture

Data Acquisition and Fuzzification

The flow begins by extracting vehicle counts from CCTV streams. In reality, these inputs are imprecise due to tropical rainstorms blur the lens and heat haze distorts the pixel matrix. The system maps each crisp vehicle onto three fuzzy membership categories into Low, Moderate and Heavy. For instance, on a scale of 0 to 100 vehicles and 45 cars are detected, the output produce the membership set {Heavy: 0.75, Moderate: 0.25}.

Rule Evaluation with Confidence Weights

The inference engine then loads conditional IF-THEN rules from the knowledge base such as IF Heavy THEN extend green by 20s. Each rule carries a confidence score that acts as a weight. Clean data receives full weight (1.00), noisy data receives reduced weight (0.50) and offline sensors receive the lowest weight (0.30). The final activation strength is the fuzzy membership value by its confidence weight. Then, R1 Heavy at $0.75 \times 0.50 = 0.375$ and R2 Moderate at $0.25 \times 0.50 = 0.125$.

Aggregation and Defuzzification

Continuing the calculation, 0.375 for R1 Heavy at 20s and 0.125 for R2 Moderate at 8s. Min-max composition aggregates the weighted rules and serves as the input for the defuzzification stage. The CoG method defuzzifies the result into a single timing decision. Here, $\Sigma(\text{firing strength} \times \text{output value}) / \Sigma(\text{firing strength})$. For example, $(0.375 \times 20 + 0.125 \times 8) / (0.375 + 0.125) = 8.5 / 0.5 = 17$ seconds. The final output, extend green phase by 17 seconds then dispatched to the relevant traffic signal.

1.2.3 Justification of Selected Method

Below shows the critical reasons to justify the suitability of the Mamdani-type fuzzy logic model to the intelligent traffic management context.

Real-Time Computational Efficiency

Unlike Bayesian networks that rely on intensive floating-point matrix calculations from large training datasets, while a pre-compiled fuzzy rule can be evaluated based on simple min-max logical operations to the values between 0.0 and 1.0. This requires minimal computation overhead as execution takes less than a millisecond (Vlasceanu, I.-M.; Ceapa, V.-C.-D.; Sacala, I.S.; Caruntu, C.F.; Udrea, A.-I.; Constantin, N. ; Segarceanu, M., 2025). This low computational rate is suitable for the real-time traffic control application, where the MBPP system may adjust the traffic lights as the traffic flow on the street constantly changes.

Handling Non-Linear Complexity

Penang often have narrower streets compared to modern uniformly planned cities. One double-parked vehicle or a group of tourist's trishaws stopped can cause enormous non-linear traffic cascades difficulty to model exactly by a mathematical formula. Fuzzy logic does not require perfect mathematical modelling, but it relies on flexible and qualitative heuristic rules that maintain optimal traffic flow even during incomplete and non-conforming traffic streams.

Elimination of Signal Chattering

Boolean reasoning would cause the automated system to switch traffic light signals back and forth based on rapid fluctuations between 39 and 41 vehicles from CCTV's input that is sensitive to pixel jitter or rain blur. This unpredictable switching is known as signal chattering. If using overlapping fuzzy membership boundaries, the minor difference in the input data causes the gradual and stable shifts of phases, thereby preventing mechanical wear from excessive operation of physical street hardware (Fahrunnisa, Z.; Rahmadwati; Setyawan, R.A., 2024).

Rule-Based Knowledge Base Integration

Mamdani fuzzy logic uses a rule-based knowledge base that represents traffic control expertise as intuitive IF-THEN rules. It is fully transparent and auditable decision-making allow traffic engineers to inspect or modify rules without retraining the system. Therefore, such explainability is particularly important for safety-critical traffic signal control.

1.3 Roles of Knowledge Bases

1.3.1 Knowledge Storage

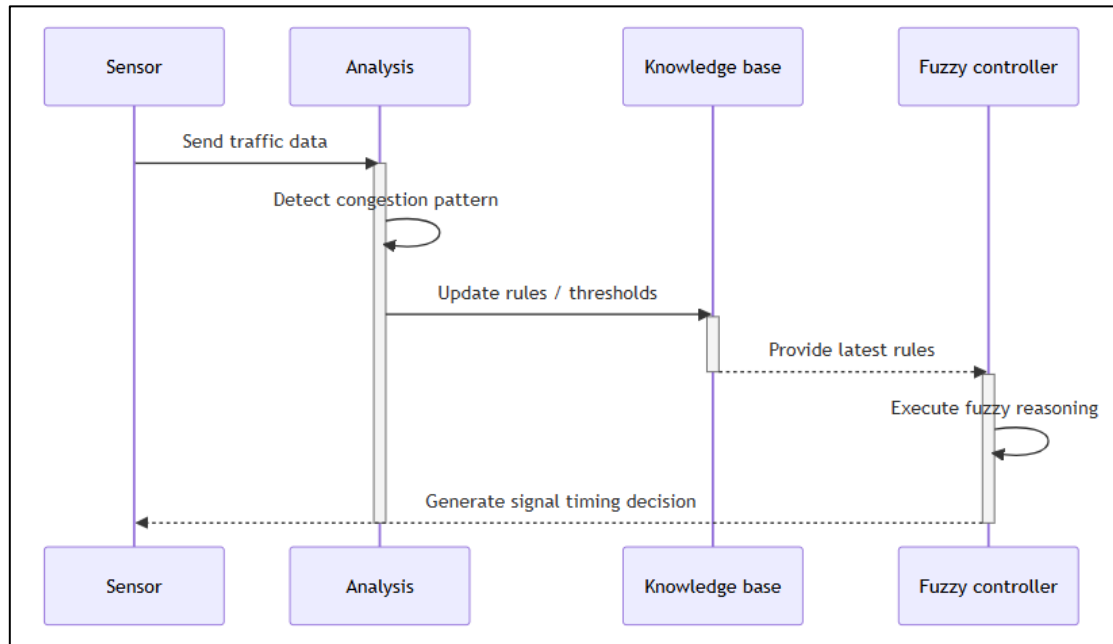


Diagram 3. Intelligent Traffic Congestion Detection Tree-Sequence UML

The knowledge base is the intelligent data repository upon, which system reasoning is based. The knowledge base contains two separate logical domains for historical stability and real-time responsiveness as mentioned below.

Static Knowledge Layer

Structural rules and geographical constants regarding municipal territory that do not change over long periods of time are stored in the static knowledge layer (Nigam, N.; Singh, D.P.; Choudhary, J., 2023). Examples include digital GIS topology maps of the urban street network, fixed lane direction matrices, legal speed limits, maximum vehicle capacity per road segment and historic baseline curves containing expected vehicle counts for any given time and day.

Dynamic Knowledge Layer

In the dynamic knowledge layer, these domains are a continuous shifting register of the current state of the physical world. This can include real-time vehicle queue counts from current CCTV frames, counts of active traffic light phases, real-time data confidence weights from the validation filter and active incident flags.

Therefore, having two distinct knowledge bases ensures that the operation remains unaffected if the incoming street data is incomplete.

1.3.2 Knowledge Update

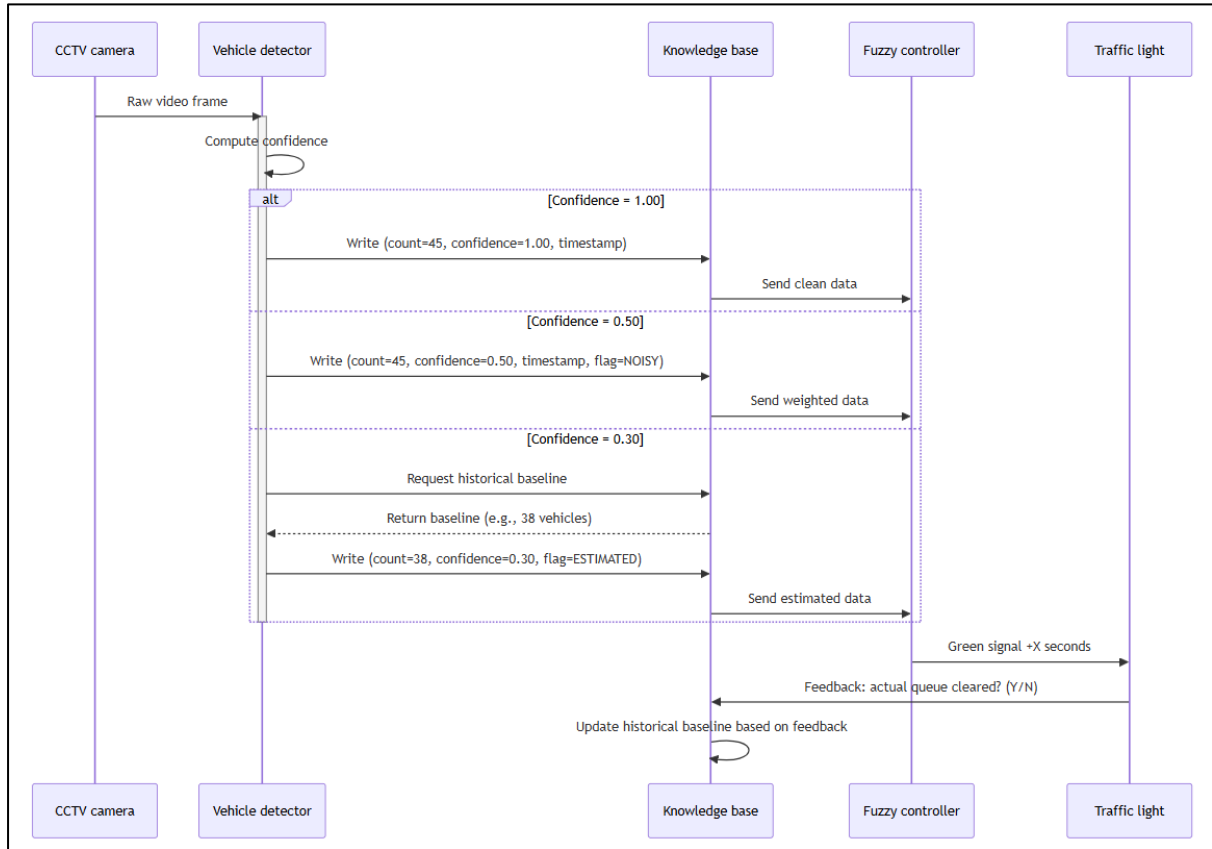


Diagram 4. Confidence-Based Traffic Control Tree-Sequence UML

Diagram 4 shows the knowledge base runs in an asynchronous processing loop to maintain system responsiveness. When new data arrives, the system checks data quality, handles missing or incomplete data and writes a timestamped state vector to the dynamic knowledge layer.

Quality Assessment and Confidence Scoring

Each CCTV frame is assigned a confidence score (1.00, 0.50 or 0.30) based on environmental conditions. As identified in Section 1.1, tropical heat haze and monsoon downpours degrade image quality in ways that are invisible to a naive system. Signal timing became erratic without confidence scoring as rain-blurred frame showing 45 vehicles would be treated identically to a clear frame showing 45 vehicles. Confidence scoring makes data reliability explicit before any further processing.

Three-tier Update Logic

The update logic operates across three confidence tiers, each handling the incoming reading differently according to the extent to which it can be trusted. When data is clean (confidence = 1.00), the live vehicle count is written directly with a `CLEAN` flag as the input accurately reflects physical reality and can be fully trusted.

During monsoon rain and heat haze, the data becomes noisy (confidence = 0.50) as the weather introduces computational noise that isn't always eliminated. The reading is retained with a `NOISY` flag and reduced weighting and signals the fuzzy inference engine to apply more conservative rule weights.

When data is offline (confidence = 0.30), the live count is unusable. The system requests a historical baseline from the static knowledge layer and writes that value with an `ESTIMATED` flag. These baseline curves that indexed by time slot ensure the fuzzy controller always receives a plausible input rather than an empty slot.

Timestamping Every Update

Every write to the dynamic layer includes the current system time. The system cannot distinguish between fresh data and old data without timestamps. A sensor that went offline 3 hours ago would still appear to have contemporary data. Timestamps enable the system to reject entries older than a threshold and trigger fallback logic. This addresses the incomplete data challenge from 1.1.

1.3.3 Criticality to System Performance

The Knowledge Base serves as a crucial failure defence in case the real-world data is incomplete or noisy. In a smart city deployment, packet loss, sensor disconnection or visual disruptions often occur.

If the camera had a downtime error, this leaves an empty data slot. The Fuzzy Logic Controller had no membership value to fuzzified. The calculation either divides by zero during defuzzification or propagates a null that crashes the control loop. If data is present but unstable, uncertainty may occur and cause the controller to oscillate its phase decisions as the signal-chattering failure mode that drives mechanical wear on physical hardware. In both cases, the fault is not the reasoning engine but the absence of trustworthy input.

The Knowledge Base resolves this by maintaining historical traffic baselines. If live data confidence drops to the lowest level, the system will automatically replace the historical baseline curve held in the static layer for the unusable live count. For example, a camera blinded by the monsoon rain sees 45 cars at a confidence weight of 0.30. The system executes the fallback and pulls 38 vehicles from the Knowledge Base. Fuzzification generates {Heavy: 0.20, Moderate: 0.80, Low: 0.00} to produce a stable green extension of approximately 12 seconds instead of system failure.

This two-layer design guarantees uninterrupted and optimised signal timing even during hardware interruptions. The knowledge layer provides a reliable foundation for urban edge networks, therefore allowing the system to continue operation without data fragmentation.

(1727 words)

2.0 Machine Learning

2.1 Choice of ML Model

A machine learning solution is required for the vehicle detection and counting module, where CCTV footage must be converted into accurate traffic-flow information despite varying weather, lighting and occlusion conditions.

The primary challenge is the unreliability of vehicle counts generated from CCTV footage during Malaysia's tropical weather. Monsoon rain, heat haze, poor lighting and motion blur introduce visual noise that causes traditional rule-based image-processing techniques to produce unstable results as shown in Figure 4. These disturbances may lead to inaccurate vehicle counts, which ultimately reduce the accuracy of traffic congestion assessment. Therefore, traditional rule-based image processing is not suitable for this purpose.

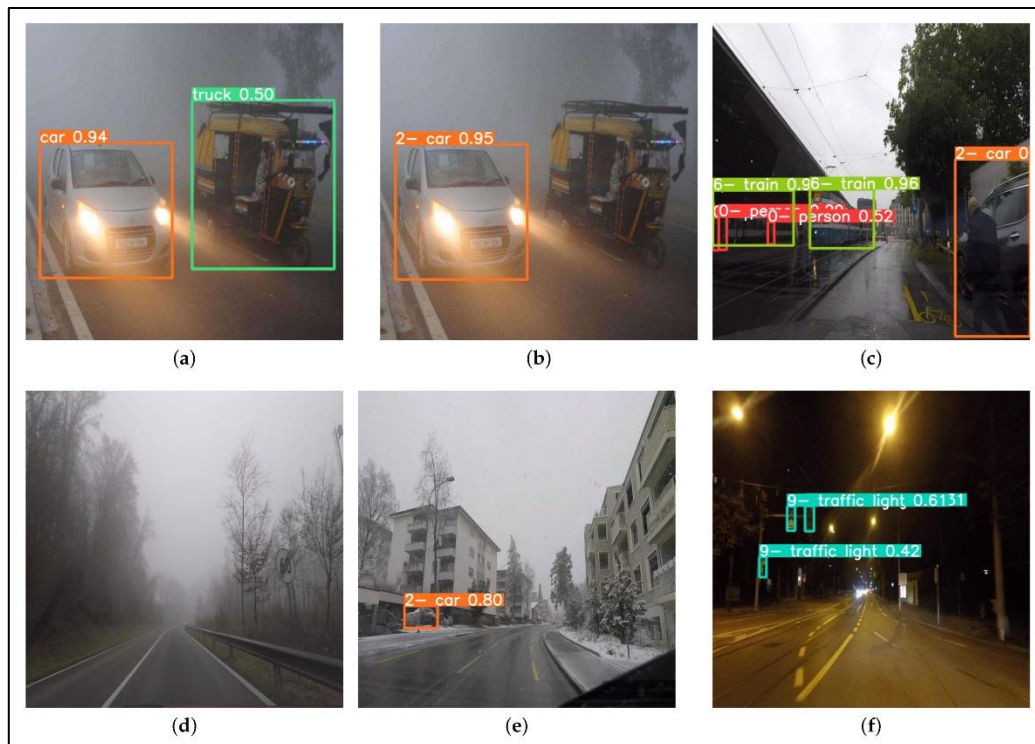


Figure 4. Detection Under Degraded Weather and Low-visibility Conditions

(Ashkanani, M.; AlAjmi, A.; Alhayyan, A.; Esmael, Z.; AlBedaiwi, M.; Nadeem, M., 2025)

2.1.1 Justification of CNN-YOLOv8

CNN integrated with the YOLOv8 object detection framework is selected. CNNs are appropriate to address the identified problem as the model is able to extract hierarchical visual features directly from image data rather than relying on manually defined edges or thresholds. The network can be trained to identify various features of vehicles even if the vision is partially obstructed by rain, haze or shadows, hence making the model robust to environmental variations. This is a critical requirement for Penang's climate.

The integration of YOLOv8 further enhances system performance by enabling real-time object detection. Unlike multi-stage detection methods, YOLOv8 processes images in a single forward pass by achieving processing speeds of around 30 to 100 FPS on edge-computing equipment (ForaSoft, 2025) (Rey, L.; Bernardos, A. M.; Dobrzycki, A. D.; Carramiñana, D.; Bergesio, L.; Besada, J. A.; Casar, J. R., 2025). This meets the sub-second response time for adaptive traffic signal control, where fast detection and decision-making are critical.

Furthermore, YOLOv8 takes advantage of transfer learning through pre-trained weights from large-scale datasets. This ensures high detection accuracy with significantly less task-specific data and reduces the cost and effort of developing a Penang-specific traffic monitoring system. Fine-tuning the model based on local footage captured under rainy, hazy, nighttime and daytime conditions allows it to adapt to Penang's unique environment.

While collecting and annotating thousands of traffic images is resource-intensive and training requires expensive GPU acceleration, these challenges are mitigated by deploying a lightweight YOLOv8-Tiny architecture to reduce both computational and data requirements.

2.2 Training Considerations

2.2.1 Environment Setup in Experiment

The experimental design was carefully structured to optimize the model for real-world traffic variations. All experiments used the spiral dataset with a 50% training-to-test ratio and a noise level of 20 to simulate the visual disturbances expected from CCTV footage. The network architecture consisted of four hidden layers with decreasing neuron configurations of 8, 8, 6 and 4 neurons. Each experiment was initialised by resetting the network weights and run for 500-1000 epochs while loss behaviour was monitored.

Experiment 1: Learning Rate

Learning Rate	Training Loss	Test Loss	Behaviour Observed
0.1	0.014	0.055	Fast but volatile, chaotic loss spikes before settling and a geometric spiral boundary.
0.03	0.013	0.058	Smooth and stable. Had a clean concentric spiral arm.
0.001	0.481	0.483	Underfitted, faint linear gradient and too weak to form a spiral shape.

Table 1. Comparison of Experimental Learning Rates

The learning rate parameter controls the step size during weight updates. The three tested values are summarised in Table 1 and the corresponding spiral-boundary behaviour is shown in Appendix A.

At 0.001, the model produced stagnant training behaviour and severe underfitting. At 0.1, convergence was rapid with the lowest absolute test loss. However, training exhibited volatile loss spikes and uncertain oscillations. In real-time traffic management, such instability would cause traffic signals to oscillate unpredictably between green phases. Conversely, 0.03 achieved the optimal balance, although it required a longer convergence duration, it eliminated oscillations and produced continuous loss reduction. Therefore, 0.03 was selected as it demonstrated the most reliable convergence without erratic fluctuations.

Experiment 2: Regularisation Rate

L2 Regularisation Rate	Training Loss	Test Loss	Generalization Gap	Behaviour Observed
0	0.021	0.103	0.082	Overfitted and had irregularly shaped boundaries that fit all noisy data points.
0.001	0.045	0.098	0.053	Clean concentric spiral boundaries and apply well to the test split.
0.1	0.5	0.5	0	Data underfitting, weights collapsed toward zero and no meaningful boundary formed.

Table 2. Comparison of Regularisation Rates

This parameter prevents overfitting by penalising large weights with the tested rates summarised in Table 2 and the decision-boundary plots provided in Appendix B.

In the absence of regularisation, the model learned noise-specific patterns and achieved near-zero training loss but failed to generalise, which is equivalent to learning Georgetown’s congestion patterns while performing poorly at Bayan Lepas or Air Itam. At 0.1, excessive penalisation drove weights toward zero, which resulted in underfitting, where the model produced near-random signal timing decisions. The optimal balance at 0.001 indicated effective suppression of noise-induced variance while preserving the genuine spiral structure. Accordingly, 0.001 was selected for subsequent experiments.

Experiment 3: Activation Function

Activation Function	Training Loss	Test Loss	Behaviour Observed
Sigmoid	0.5	0.5	Total learning failure and blank boundary canvas.
Linear	0.467	0.475	Underfitted and flat diagonal split.
ReLU	0.021	0.070	Optimal and geometric spiral arms are sharp and tightly bound.

Table 3. Comparison of Neural Network Activation Function Types

This function introduces non-linearity to allow the model to learn complex patterns. From Table 3, the Linear activation function failed to learn the spiral's curved decision boundaries. Sigmoid performed worse as error signals diminished exponentially across the four hidden layers. As a result, no significant weight updates are produced. The inability to distinguish between moderate and severe congestion levels provides no reliable basis for signal timing decisions. ReLU overcame this by maintaining positive gradients for all active neurons, thereby ensuring effective backpropagation through all layers. Hence, ReLU was selected.

2.2.2 Risks of Poor Parameter Choices

Poor hyperparameter selection can degrade both the accuracy and reliability of the intelligent traffic management system. Since the model's vehicle counts directly inform traffic signal decisions, unstable training behaviour may lead to inaccurate signal adjustments.

An excessively high learning rate may cause inconsistent predictions of green-phase durations produced. Conversely, an excessively low learning rate slows optimisation, thereby requiring longer training times and slower adaptation to traffic changes (Goodfellow, I.; Bengio, Y.; Courville, A., 2016).

A regularisation rate of 0 causes the model to memorise instead of learning general patterns, whereas an excessively high regularisation rate prevents the model from learning traffic characteristics (Goodfellow, I.; Bengio, Y.; Courville, A., 2016). Both extreme rates perform poorly in traffic.

For activation functions, Sigmoid suffers from slow learning and reduces training efficiency, while Linear activation is unable to capture non-linear traffic patterns. As a result, the model fails to respond appropriately to changing conditions and generates oversimplified signal timing recommendations.

2.3 Human-AI Interaction

2.3.1 System Communication Results to Users

The traffic management system communicates its real-time decisions through a centralised dashboard at the MBPP IOC. The dashboard displays the current vehicle count, congestion level, recommended green-time extension and a live CCTV feed. Traffic operators are allowed to verify the detected traffic conditions before taking action. This ensures that human oversight remains an essential part of the decision-making process, particularly in complex urban traffic environments, where external factors may influence traffic flow.

Explainable AI (XAI) features

Recommendations are accompanied by confidence indicators using a three-tier colour scheme, where green represents verified clean data, amber indicates noisy but usable data and red denotes estimated fallback data. This enables operators to quickly assess the reliability of each recommendation and determine whether manual intervention is required.

Additionally, operators can access the reasoning behind every decision, including the activated fuzzy rules, membership values and the final defuzzified output. The information is presented in an understandable format to help operators interpret system behaviour and increase trust in AI-assisted decision-making (Agostinho, C.; Dikopoulou, Z.; Lavasa, E.; Perakis, K.; Pitsios, S.; Branco, R.; Reji, S.; Hetterich, J.; Biliri, E.; Lampathaki, F.; Rodríguez Del Rey, S.; Gkolemis, V., 2023). This traceability also supports accountability as every recommendation can be audited and reviewed if unexpected traffic outcomes occur.

Real-time alerts are delivered during abnormal situations, such as sensor failures, prolonged use of historical data or activation of the safety control mode. Each alert identifies the affected junction, explains the issue and recommends an appropriate action, thereby enabling operators to respond promptly and minimise traffic congestion.

2.3.2 Improving System Performance Through User Feedback

The system employs a human-in-the-loop feedback mechanism to continuously learn and adapt to changing urban traffic dynamics. Operator feedback is treated as valuable training data for iterative improvement that is gathered through different channels.

Operator overrides occur when an operator identifies an anomaly that AI missed such as funeral processions, VIP escort or a rain-blocked camera lens. The operator manually overrides the traffic light timings and the system logs this intervention as an error correction. These overrides are periodically reviewed to retrain the model on similar edge cases in future deployments.

Overriding logging activates automatically whenever an operator manually changes signal timing. The system records the adjustment alongside its own original recommendation. If the same type of override occurs repeatedly under similar traffic conditions, the ML component automatically adjusts the confidence weights of the relevant fuzzy logic rules (Pereira, J. C. G.; Domingos, G.; Rosa, L. G., 2025).

Ground truth reporting occurs when an operator manually inputs observed traffic conditions that differ from the system's detection. For instance, an operator reporting "Light" traffic when the system detected "Moderate" due to a blurry camera. This feedback retrains the confidence weighting mechanism to reduce the influence of unreliable inputs.

2.3.3 Risks of Over-Trust and Under-Trust

Risk of Over-Trust

Automation complacency occurs when operators stop verifying outputs and defer entirely to automated decisions. A prominent risk occurred during automated infrastructure control framework trials, where a critical software glitch simultaneously froze major intersections at red lights. As operators had over-trusted the automation, it led to a delay of manual intervention by turning an isolated system bug into a regional gridlock (Yu, H.; Hulse, D.; Irshad, L., 2024).

In the context of the proposed system, over-trust can cause an operator to reflexively trust corrupted vehicle counts during severe monsoons and blindly trust confidence-tier alerts. This automation bias results in the fallback of knowledge-based systems to run indefinitely without investigating the underlying sensor failure.

Risk of Under-Trust

Conversely, under-trust occurs when operators are resistant to algorithmic choices due to a cognitive bias toward human intuition. Recent studies indicate that traffic police frequently disable automated systems during peak hours by switching to manual control (Vajeeran, A.; De Silva, D., 2022). Post-deployment studies demonstrated that manual control resulted in worse congestion, given that human operators cannot calculate network-wide traffic flows as comprehensively as a computer.

According to traffic micro-simulation research, network-wide optimisation models reduce overall delays by 43.5% and queue lengths by 61.9% compared to uncoordinated control styles (Srisurin, P.; Guerra, A.; Jarumaneeroj, P., 2024). Human operators can only process visible local queues causes long manual cycle choices taken to break multi-intersection green waves. Severe queue spillbacks on minor approaches and magnifying total network-wide gridlock are generated instantly.

2.4 Importance of the Machine Learning Component

The absence of the ML component would render the entire system ineffective at converting raw CCTV footage into reliable vehicle counts. The system would fail in the following ways.

Perception Failure

Traditional image processing techniques such as edge detection or background subtraction fail under Penang's tropical heat haze and monsoon rain as identified in Section 1.1. In the absence of YOLOv8's confidence scores, a rain-blurred frame showing 45 vehicles would be treated identically to a clear frame. This causes the fuzzy logic controller to apply full rule weights to unreliable data and produce erratic green-time extensions that could worsen congestion.

Resilience Failure

When a camera goes offline due to network failure or physical damage, the ML component's confidence scoring triggers a fallback to historical baselines stored in the static knowledge base layer. In the absence of ML, the system would receive NULL values or crash entirely and the knowledge base would remain empty, which leaves the system with no memory of typical traffic patterns for any given time or day.

Adaptation Failure

Human overrides cannot be converted into structured retraining data without an ML component and unable to convert into structured retraining data. Operator interventions would remain isolated incidents causes the system to never learn from repeated errors. Over time, this accumulation of unaddressed mistakes would demand constant manual intervention across dozens of intersections and lead to system unsustainable.

Scalability Failure

Georgetown covers 113 square miles with dozens of major intersections. Therefore, manual vehicle counting impossible at this scale. The absence of ML's real-time detection capability prevents the system from processing multiple CCTV feeds simultaneously, thereby creating critical blind spots across the road network. The lack of YOLOv8's vehicle classification prevents the system from differentiating vehicle types affecting the accuracy of prioritising public transport or accurately detecting the motorcycles that dominate Penang's roads (Salih, A. B.; Mohammed, R. B.; Othman, A. A.; Mostafa, S. A.; Kamalrudin, M., 2026).

(1759 words)

3.0 References

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4.0 Appendix

Appendix A: Comparison of Experimental Learning Rates

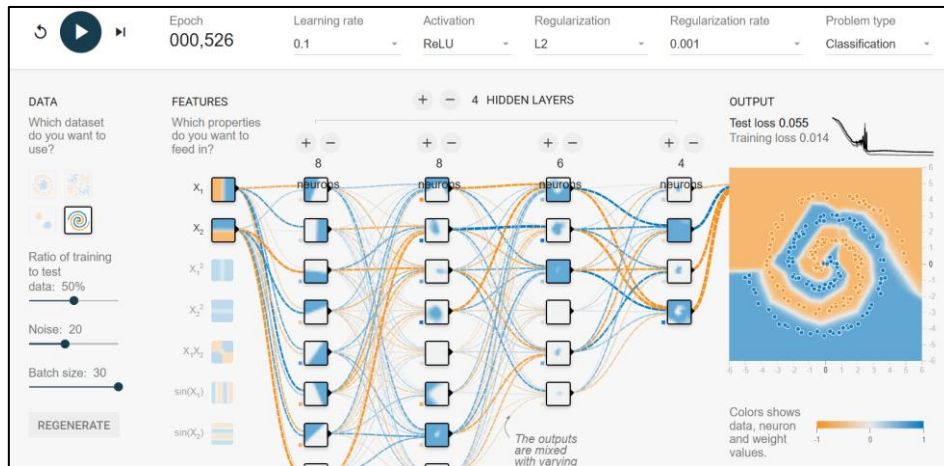


Figure A.1 Experimental Results for Learning Rate = 0.1

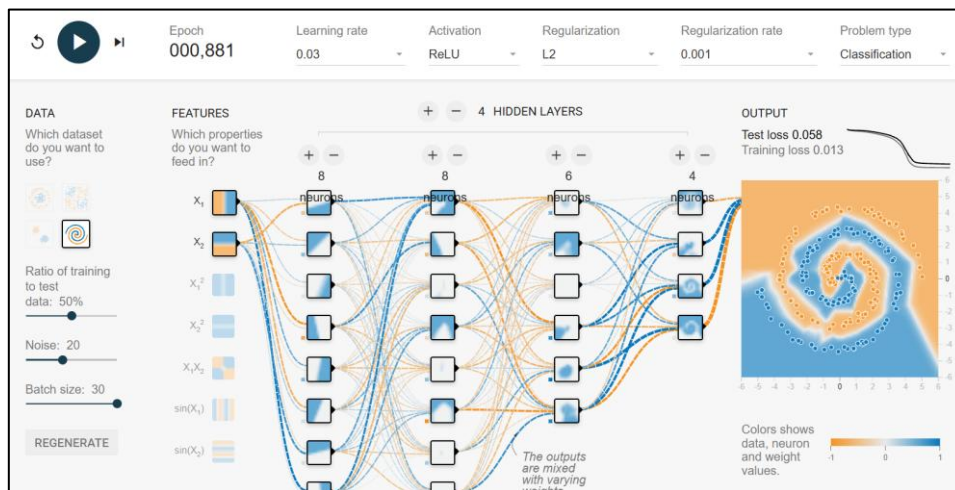


Figure A.2 Experimental Results for Learning Rate = 0.03

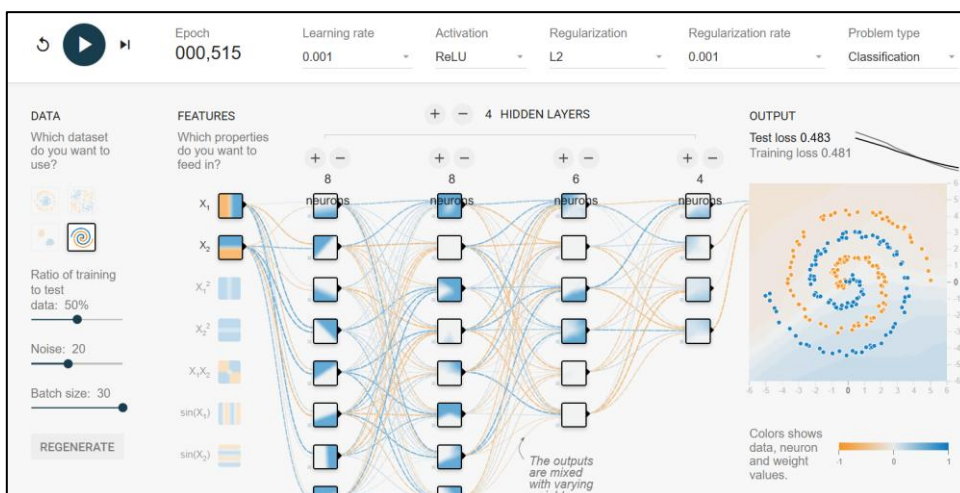


Figure A.3 Experimental Results for Learning Rate = 0.001

Appendix B: Comparison of Regularisation Rates

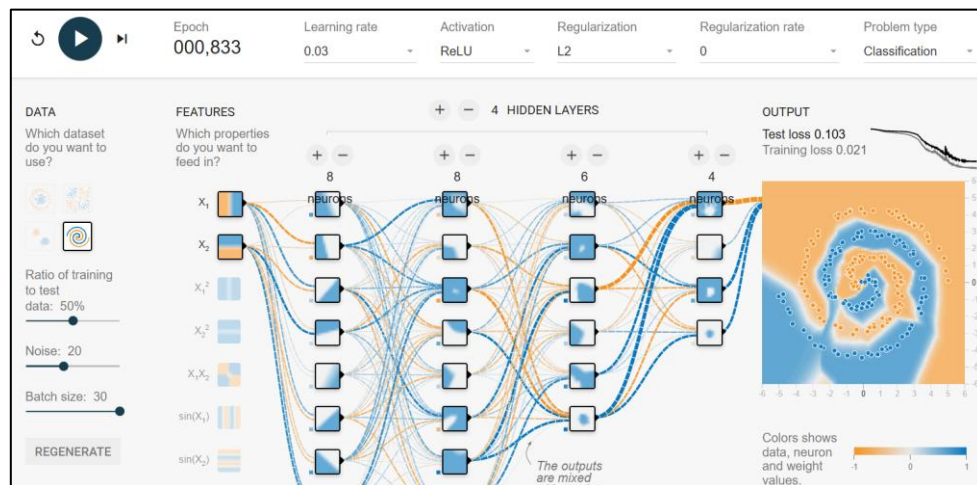


Figure B.1 Experimental Results for Regularisation Rate = 0

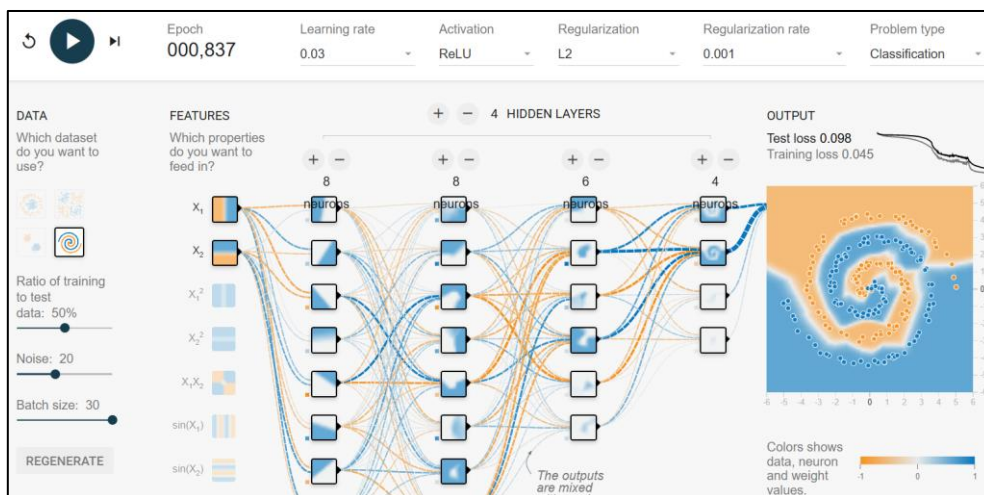


Figure B.2 Experimental Results for Regularisation Rate = 0.001



Figure B.3 Experimental Results for Regularisation Rate = 0.1

Appendix C: Neural Network Activation Functions

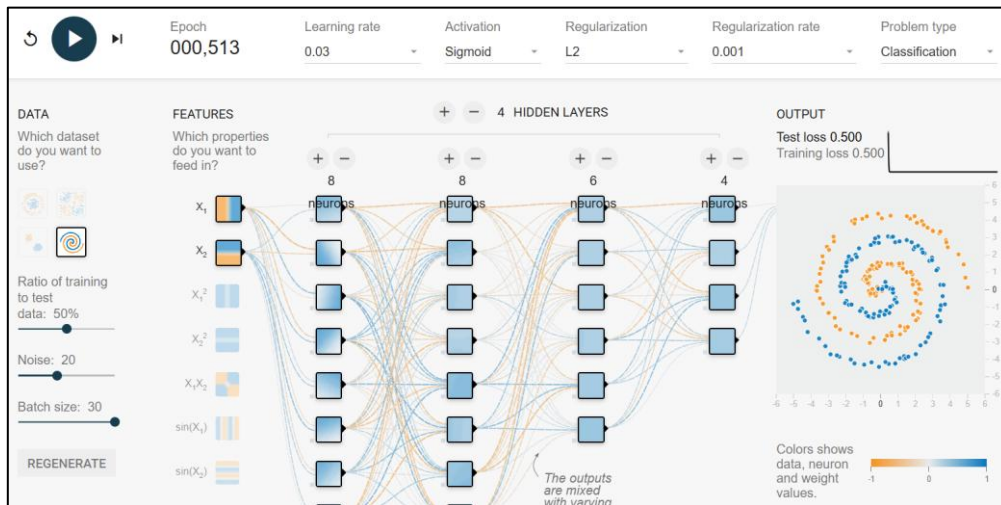


Figure C.1 Sigmoid Activation Function Graph

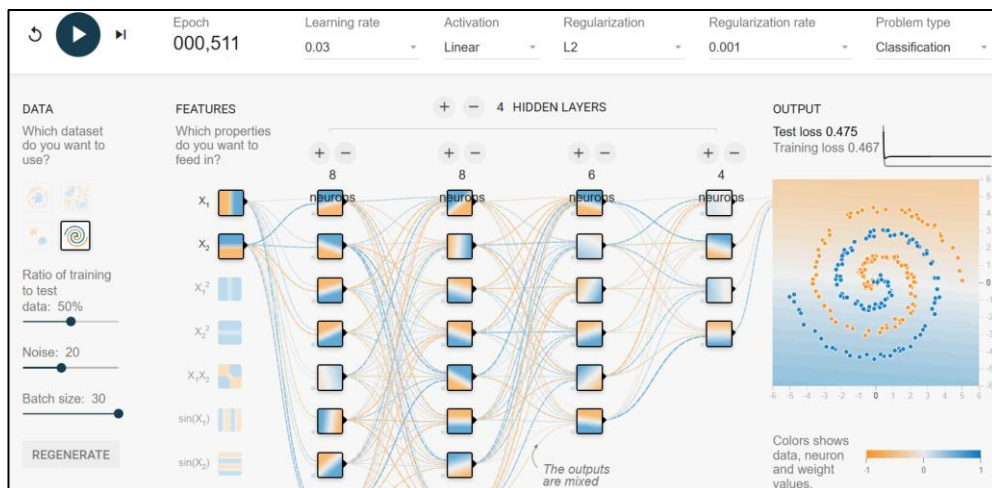


Figure C.2 Linear Activation Function Graph

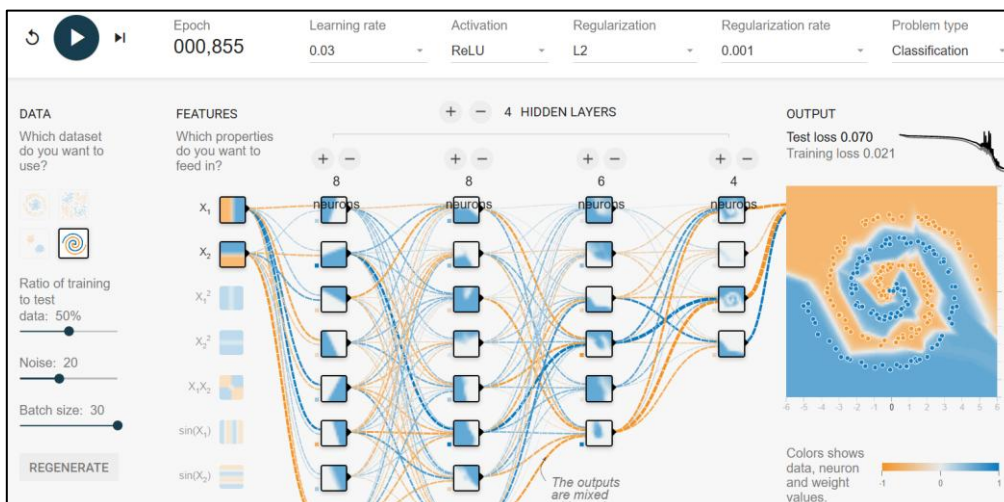


Figure C.3 ReLU Activation Function Graph

Appendix D: Word Count

My Writing Details	
Words	1,727
Characters	11,602
Sentences	91
Paragraphs	26

Task 1 Word Count

My Writing Details	
Words	1,759
Characters	12,725
Sentences	95
Paragraphs	33

Task 2 Word Count

Appendix E: Turnitin Plagiarism Checker

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